

Time Series Analysis

Chapter 5 – ARMA models

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Chapter 1

ARMA models

Introduction

Chapter ?? introduced stationarity and the sample ACF/PACF; Chapter ?? showed how to turn a non-stationary series into a stationary one by differencing. This chapter assembles the central modelling family for stationary time series: the *autoregressive moving-average* processes, $\text{ARMA}(p, q)$. We give their precise definitions, state the stationarity and invertibility conditions in terms of polynomial roots, derive their ACF/PACF *signatures*, and detail the practical workflow — the Box–Jenkins method — for identification, estimation, diagnostics and forecasting.

1.1 Autoregressive process $\text{AR}(p)$

Definition 1 – Autoregressive process $\text{AR}(p)$

A process $(X_t)_{t \in \mathbb{Z}}$ is autoregressive of order p if

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2),$$

with $\phi_p \neq 0$. Equivalently, with $\Phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$,

$$\Phi(B) X_t = \varepsilon_t.$$

Proposition 2 – Stationarity of $\text{AR}(p)$

Equation $\Phi(B) X_t = \varepsilon_t$ admits a unique causal stationary solution if and only if every root z_i of $\Phi(z) = 0$ satisfies $|z_i| > 1$. In that case

$$X_t = \Phi^{-1}(B) \varepsilon_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j},$$

with $\sum_j |\psi_j| < \infty$.

Example 3 – $\text{AR}(1)$ moments

For $X_t = \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$, iterating gives $X_t = \sum_{j \geq 0} \phi^j \varepsilon_{t-j}$, hence

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \frac{\sigma^2}{1 - \phi^2}, \quad \rho(h) = \phi^{|h|}.$$

The ACF **decays geometrically** and the PACF **cuts off** after lag 1 ($\alpha(1) = \phi$, $\alpha(h) = 0$ for $h \geq 2$).

Proposition 4 – Yule–Walker equations

Let $X_t \sim \text{AR}(p)$ be stationary. Multiplying the equation by X_{t-h} and taking expectations gives, for $h = 1, \dots, p$,

$$\rho(h) = \phi_1 \rho(h-1) + \phi_2 \rho(h-2) + \dots + \phi_p \rho(h-p).$$

In matrix form $\mathbf{R}\phi = \mathbf{r}$, the system is invertible under the stationarity condition; replacing ρ by the sample ACF $\hat{\rho}$ gives the Yule–Walker estimator $\hat{\phi}_{\text{YW}}$.

Example 5 – AR(2) stationarity triangle

For $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$, the roots of $1 - \phi_1 z - \phi_2 z^2 = 0$ lie outside the unit disk iff

$$\phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1, \quad -1 < \phi_2 < 1.$$

This is the Schur stability condition for the AR(2) polynomial.

1.2 Moving-average process MA(q)

Definition 6 – Moving-average process MA(q)

A process $(X_t)_{t \in \mathbb{Z}}$ is moving-average of order q if

$$X_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2),$$

with $\theta_q \neq 0$. Equivalently, $X_t = \Theta(B) \varepsilon_t$ with $\Theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$.

Proposition 7 – Moments and ACF of MA(q)

Every MA(q) is (weakly) stationary, with

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \sigma^2(1 + \theta_1^2 + \dots + \theta_q^2),$$

and autocovariance

$$\gamma(h) = \begin{cases} \sigma^2(\theta_h + \theta_1 \theta_{h+1} + \dots + \theta_{q-h} \theta_q) & 1 \leq h \leq q, \\ 0 & h > q, \end{cases}$$

with $\theta_0 = 1$. In particular $\rho(h) = 0$ for $h > q$: the ACF **cuts off** at lag q .

Definition 8 – Invertibility

An MA(q) is *invertible* if it admits an AR(∞) representation

$$\varepsilon_t = X_t - \pi_1 X_{t-1} - \pi_2 X_{t-2} - \dots, \quad \sum_j |\pi_j| < \infty.$$

This holds if and only if every root of $\Theta(z) = 0$ lies *outside* the closed unit disk.

Remark – Why invertibility matters

Without invertibility, two parameter sets give the same ACF and so are indistinguishable from the data. The MA(1) models $X_t = \varepsilon_t + \theta\varepsilon_{t-1}$ and $X_t = \tilde{\varepsilon}_t + (1/\theta)\tilde{\varepsilon}_{t-1}$ have identical autocovariance functions; we conventionally choose the invertible one, $|\theta| < 1$.

1.3 ARMA(p, q)**Definition 9 – ARMA(p, q)**

A process (X_t) is ARMA(p, q) if it satisfies

$$\Phi(B)X_t = \Theta(B)\varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2),$$

with $\Phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$, $\Theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$, and Φ, Θ sharing no common root.

Proposition 10 – Stationarity, invertibility and existence

The equation $\Phi(B)X_t = \Theta(B)\varepsilon_t$ admits a unique causal stationary invertible solution if and only if

$$|z_i| > 1 \text{ for every root of } \Phi, \quad |z_j| > 1 \text{ for every root of } \Theta.$$

The solution is then

$$X_t = \Phi^{-1}(B)\Theta(B)\varepsilon_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}.$$

Remark

The MA(∞) coefficients (ψ_j) are obtained by polynomial division of $\Theta(z)$ by $\Phi(z)$ around $z = 0$. They control both the autocovariance and the forecast variance (Section 1.8).

1.4 Box–Jenkins identification

The choice of (p, q) is guided by the sample ACF and PACF, whose *signatures* differ across model classes:

Model	ACF $\rho(h)$	PACF $\alpha(h)$
AR(p)	exponential decay	cuts off after lag p
MA(q)	cuts off after lag q	exponential decay
ARMA(p, q)	exponential decay (after q)	exponential decay (after p)

Remark – Bartlett bands

“Cuts off” is read as $\hat{\rho}(h)$ falling *inside* the $\pm 1.96/\sqrt{T}$ band for h larger than the cutoff lag. “Decays” means the values stay outside the band but shrink in absolute value.

Example 11 – Reading a sample ACF/PACF

On a series of length $T = 200$, the sample PACF shows $\hat{\alpha}(1) = 0.62$, $\hat{\alpha}(2) = -0.04$, $\hat{\alpha}(3) = 0.02$ and the sample ACF decays geometrically. The Bartlett band is $\pm 1.96/\sqrt{200} \approx \pm 0.139$, so $\hat{\alpha}(h)$ is inside the band for $h \geq 2$: the PACF cuts off at lag 1, suggesting AR(1).

1.5 Estimation**1.5.1 Yule–Walker (AR only)**

For pure AR(p), plug $\hat{\rho}$ into Proposition 1.1 to obtain the Yule–Walker estimate. This is method-of-moments: *consistent* but not asymptotically efficient.

1.5.2 Conditional sum of squares (CSS)

Conditioning on the first p observations and assuming $\varepsilon_t = 0$ for $t \leq 0$, the CSS estimator is

$$\hat{\theta}_{\text{CSS}} = \arg \min_{\theta} \sum_{t=p+1}^T \hat{\varepsilon}_t(\theta)^2,$$

where $\hat{\varepsilon}_t$ is computed recursively from the ARMA equation. CSS is fast and equivalent to MLE in large samples under Gaussian innovations.

1.5.3 Maximum likelihood (MLE)

Under $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, the joint distribution of $(X_1, \dots, X_T)$ is multivariate normal with covariance matrix $\Gamma_T(\theta)$. The log-likelihood

$$\log \mathcal{L}(\theta, \sigma^2) = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \log \det \Gamma_T(\theta) - \frac{1}{2} \mathbf{X}^\top \Gamma_T(\theta)^{-1} \mathbf{X}$$

is maximised numerically. The standard implementation in `statsmodels` uses a state-space representation and the Kalman filter, which scale linearly in T and provide asymptotic standard errors.

Theorem 12 – Asymptotic distribution of the MLE

Under regularity assumptions (stationarity, invertibility, no common roots), as $T \rightarrow \infty$,

$$\sqrt{T}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}),$$

where $I(\theta_0)$ is the Fisher information matrix evaluated at the true parameter θ_0 .

1.6 Model selection**Definition 13 – Information criteria**

For a fitted ARMA(p, q) with $k = p + q + 1$ free parameters (including σ^2),

$$\text{AIC} = -2 \log \hat{\mathcal{L}} + 2k, \quad \text{BIC} = -2 \log \hat{\mathcal{L}} + k \log T.$$

The chosen order minimises the criterion.

Remark – AIC vs BIC

AIC favours predictive accuracy and tends to over-fit on small samples; BIC penalises complexity more and is consistent for the true order under standard assumptions. When the two disagree, prefer the BIC choice for parsimony and confirm with the residual diagnostics.

1.7 Residual diagnostics**Remark – Adequacy checks**

A well-specified ARMA model produces residuals $\hat{\varepsilon}_t = X_t - \hat{X}_{t|t-1}$ that are *indistinguishable from white noise*. Standard checks:

- time plot of $\hat{\varepsilon}_t$ – mean stable around zero, variance constant;
- sample ACF of $\hat{\varepsilon}_t$ – spikes inside Bartlett bands;
- Ljung–Box test;
- QQ-plot vs $\mathcal{N}(0, 1)$ for normality (Jarque–Bera test).

Theorem 14 – Ljung–Box test on residuals

$$Q_{LB}(m) = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}_{\hat{\varepsilon}}(h)^2}{T-h}.$$

Under the null that $\hat{\varepsilon}_t$ is white noise, after fitting an ARMA(p, q),

$$Q_{LB}(m) \xrightarrow{d} \chi_{m-p-q}^2.$$

Reject the null if $Q_{LB}(m)$ exceeds the $\chi_{m-p-q, 1-\alpha}^2$ quantile.

1.8 Forecasting**Definition 15 – Best linear forecast**

The minimum mean-squared-error linear forecast at horizon h , given the past $\mathcal{F}_T = \sigma(X_T, X_{T-1}, \dots)$, is

$$\hat{X}_{T+h|T} = \mathbb{E}[X_{T+h} | \mathcal{F}_T].$$

The forecast error is $e_{T+h} = X_{T+h} - \hat{X}_{T+h|T}$.

Proposition 16 – Forecast error variance

With the MA(∞) representation $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$,

$$e_{T+h} = \sum_{j=0}^{h-1} \psi_j \varepsilon_{T+h-j},$$

hence

$$\mathbb{E}[e_{T+h}] = 0, \quad \text{Var}(e_{T+h}) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

Example 17 – Forecast for AR(1)

With $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$:

$$\hat{X}_{T+h|T} = \phi^h X_T, \quad \text{Var}(e_{T+h}) = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2}.$$

As $h \rightarrow \infty$, $\hat{X}_{T+h|T} \rightarrow 0$ (the unconditional mean) and the variance tends to $\sigma^2/(1 - \phi^2)$ (the unconditional variance).

Remark – Confidence intervals

Under Gaussian innovations, the $1 - \alpha$ confidence interval is

$$\hat{X}_{T+h|T} \pm z_{1-\alpha/2} \sigma \sqrt{\sum_{j=0}^{h-1} \psi_j^2}.$$

For stationary ARMA, the width converges to a finite limit; for ARIMA (Chapter ??), it grows without bound.

Summary

Key takeaways.

- AR(p) depends on past values; MA(q) depends on past shocks; ARMA(p, q) combines both.
- Stationarity = roots of Φ outside the unit disk; invertibility = roots of Θ outside the unit disk.
- ACF/PACF signatures: AR cuts the PACF, MA cuts the ACF, ARMA shows double exponential decay.
- Estimation: Yule–Walker for pure AR; CSS or MLE for general ARMA. `statsmodels.tsa.arima.ARIMA` provides both.
- Model selection by AIC / BIC; validate with Ljung–Box on residuals.
- Forecasts widen with horizon according to $\text{Var}(e_{T+h}) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$.